

Pharmacology (4 credits)

Fall 2024

Adjunct Faculty Instructor: Diana Kazolias, M.S., M.B.A.

diana.kazolias@unh.edu

Class meeting: Wednesday 6:10 to 9:10 pm via Zoom

Office hours: Mondays 7:10 to 8:10 pm <https://unh.zoom.us/j/7018530036> or by appointment (please email your preferred contact method when requesting time)

This introductory course in modern pharmacology is suitable for students interested in clinical medicine, biological science research, the pharmaceutical industry and development of modern therapeutic agents. The emphasis will be on the big picture of "therapeutic drugs" and to prepare students for future success in modern medicine, research, industry, or matriculation to health science programs.

This course covers different drug categories for major organ systems including the nervous system, the cardiovascular system, and the endocrine/reproductive system, as well as drugs for conditions like cancer, hypertension, and inflammatory diseases. Drugs will be studied from a systemic approach emphasizing the role of drugs in cell signaling, cell receptors, and effects on biochemical pathways.

Text: Pharmacology 8e, Hitner and Nagle, McGraw-Hill, 2022 with online access.

Prerequisites: (BIOL 413 with a minimum grade of C- and BIOL 414 with a minimum grade of C-) or (BMS 507 with a minimum grade of C- and BMS 508 with a minimum grade of C-) or (BIOL 411 with a minimum grade of C- and BIOL 412 with a minimum grade of C-).

Mutual Exclusion: No credit for students who have taken BMCB 760.

Objectives:

1. The student will become familiar with the different classes of drugs currently used today.
2. The student will understand drug mechanisms and how these mechanisms are related to normative physiology and pathophysiology of the human body.
3. The student will understand how drugs are administered, distributed, and metabolized in the body.
4. The student will apply textbook and classroom learning via ongoing, interactive, online discussions of external current awareness, research, and personal perspective.

Course Logistics

- When assignments are due, they are due by 5:30p EST on the due date.
- Deadlines are an unavoidable part of being a professional and this course is no exception.
- Avoid any inclination to procrastinate.
- Late assignments or class absence may only be accepted without penalty with (1) notification Lisa Enright, Assistant Dean of Student Success or (2) by advanced permission of the instructor.

Course Schedule

DATE	TOPIC	CHAPTER(S)
Aug 28	Welcome session	
Sep 4	Introduction, pharmacokinetics, geriatrics	1, 2, 3
Sep 11	Review of autonomic nervous system	5, 6, 7, 8
Sep 18	Relaxants, anesthetics, CNS, sedatives	9, 10, 11, 12
Sep 25	Antipsychotics, antidepressants	13, 14
Oct 2	Psychotomimetics, antiepileptics, Parkinson's	15, 16, 17
Oct 9	Anesthesia and analgesia	18, 19, 20
Oct 16	Cardiac, antiarrhythmics, antianginal drugs	21, 22, 23, 24
Oct 23	Diuretics, antihypertensives, hematological agents	25, 26, 27, 30
Oct 30	Nutrition, hypolipidemias, respiratory drugs	28, 29, 31, 32
Nov 6	GI, endocrine, steroids	33, 34, 35, 36
Nov 13	Gonadal, contraceptives, ED drugs Ca++ balance	37, 38
Nov 20	Pancreas and diabetes, pituitary, antineoplastics	39, 40, 45, 46
Dec 4	Review and special topics	

See Canvas for course outlines, more grade information, and updates. Check announcements often!

Grading: 25% exams, 15% homework, 35% class participation, 15% reading reflections, 10% oral presentation

Grade scale:

94-100	A	80-83.9	B-	67-69.9	D+
90-93.9	A-	77-79.9	C+	64-66.9	D
87-89.9	B+	74-76.9	C	60-63.9	D-
84-86.9	B	70-73.9	C-	Below 60	F

Details:

This course requires access through McGraw-Hill Connect, which is accessible via the *McGraw Hill Connect* tab within Canvas. Homework assignments and tests

Each chapter has a PowerPoint file. Please read the chapters before attempting the corresponding assignments.

Each chapter has online Smartbooks and assignments accessible through your Connect site. These are open book and untimed. Please take the time to work through them as they are designed to help you learn the material. The due date for each assignment is listed on each assignment. You can complete as much or as little of an assignment in a sitting, close your work and come back to it. I recommend that you not work for more than 30 minutes without a short break. Do not wait until close to the due date of an assignment to complete it. Late assignments (both Smartbooks and assignments) receive a 10% grade penalty for each day they are late. Fifteen points will be deducted for students attempting a chapter assignment before its corresponding Smartbook.

Weekly Reading Reflections - You will be required to post via Canvas online discussion board by 5:30p on the day before class. Appropriate subject matter for the discussion board may include insights

gained from relevant news/research articles that you found on your own. Responses should not be 'off the top of your head' and must be substantive (e.g., you shouldn't just say 'me too' in response to someone else's post without adding a new thought or idea). Students unable to complete all assigned posts by the required time will get a zero for that assignment.

Oral Presentation – Each student will be given the opportunity to select a disease, disorder or condition of interest to them. The presentation will address the learning objectives of this course and include a persona representative of that patient population.

There are two exams.

- The first exam (midterm) will be an open book written exam with a window of approximately five days to complete.
- The second exam (final) will include 60 questions accessible through Connect. When you log onto the exam, you will have 100 minutes to complete it. This will be a closed book, timed exam. No credit for work completed after the time is up.

Academic Honesty

The Biology Program at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<http://www.unh.edu/vpsas/handbook/academic-honesty>). By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual assignments or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report. Cheating is mainly concerned with copying on exams or in lab, bringing crib notes into an exam or referring to notes, or textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and put away for the duration of the exam. Ear buds are not permitted. Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Netiquette Guidelines

Netiquette is the socially and professionally acceptable way to communicate on the Internet. Please abide by these guidelines of "netiquette" when using online communication tools with your classmates and instructor.

In Zoom or video conferences: Turn on your camera when your network bandwidth and learning space allows. Facial expressions and body language are an important part of communicating. Connect a few minutes early. Avoid background noise. Consider using a headset to reduce distractions. Keep your device (phone, computer, etc.) on mute unless you are speaking. Speak clearly, but not too loudly. Remember that a video conference has the same degree of respect as a live class, consider your appearance behavior on camera the same you would in the classroom.

In discussion boards and email messages: Identify yourself. Begin messages with a greeting and close with your name. Avoid sarcasm. It can be misinterpreted and cause hurt feelings. Keep the dialog collegial and professional. Some discussion topics may be controversial. Do not flame -These are outbursts of extreme emotion or opinion. Think twice before you submit a response. You cannot edit or delete your posts once they have been submitted. Do not use offensive language or profanity. Use clear subject lines for your posts. Do not use all caps. It is the online equivalent of YELLING! Use emoticons to clarify your emotions. They add context to your words that cannot be seen otherwise. :) Be forgiving. Anyone can make a mistake.

Center for Academic Enrichment (CAE):

The CAE is a resource for all students enrolled in UNH Manchester courses. Our mission is to help under-prepared students prepare, prepared students advance, and advanced students excel. Trained peer tutors and professional staff provide free assistance with reading and writing in all courses, as well as content in all math and science courses. Support for other content areas is dependent on the background and competencies of current peer tutor staff. *All enrolled students are entitled to 1 hour of free one-to-one tutoring per week, per course.* In addition, peer and professional tutors may run study groups for your class and provide drop-in math tutoring. Please visit us online at manchester.unh.edu/student/center-academic-enrichment to learn more about our services or to make an appointment with a tutor.

ADA Statement:

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Student Accessibility Services (SAS) office. The Student Accessibility Coordinator at UNHM is Jenessa Zurek. Please reach out to the SAS office via email at jenessa.zurek@unh.edu for registration information and disability related questions.

For Emotional and Mental Health Services: In partnership with The Mental Health Center of Greater Manchester, UNHM offers consultation visits on a walk-in basis in Room 410H (located in the Student Services suites 4th floor). Services include: Free confidential screening & consultation with a licensed mental health clinician. Referrals to mental health or substance misuse treatment, and assistance in understanding how to afford additional treatment (with or without insurance!) or find free services.

Check the fall semester walk-in hours or you may call ahead to be placed on the schedule by calling Academic Advising at (603) 641 - 4170. Or directly call The Mental Health Center of Greater Manchester at (603) 668 - 4111.

Confidentiality and Mandatory Reporting: The University of New Hampshire at Manchester and its community are committed to assuring a safe and productive educational environment for all students and for the university. To this end, the university requires UNH faculty and staff members, hall directors, resident assistants (RAs), and other students employed in leadership, supervisory, and/or mentoring roles [such as Peer Assistant Leaders (PALs), Mentor Ambassadors for Transfer Engagement (MATEs) and the Center for Academic Enrichment (CAE) mentors and tutors] to report to the university's Title IX Coordinator (Donna Marie Sorrentino, dms@unh.edu, 603-862-2930/1527 TTY) any incidents of sexual violence, domestic violence, stalking and harassment shared by students regardless of whether it occurs on or off the Manchester campus. If you wish to speak to a confidential support service provider who does not have this reporting responsibility because their discussions with clients

are subject to legal privilege, you can find a list of confidential crisis centers here. Please contact the crisis center that services the town you live in. For more information about what happens when you report, how the university considers your requests for confidentiality once a report is made to the Title IX Coordinator, your rights and report options at UNH (including anonymous report options) please visit <https://www.unh.edu/affirmativeaction/reporting-students>.

This syllabus is subject to change.